

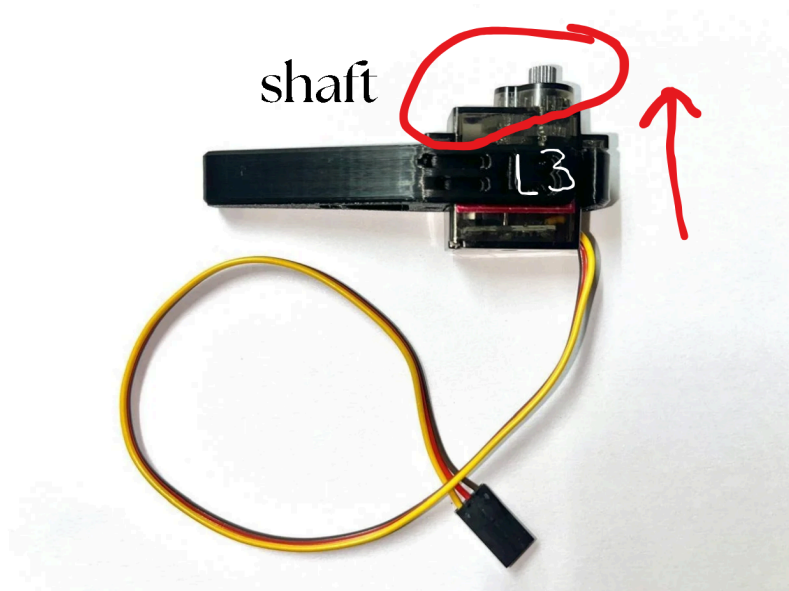
# Single Leg Assembly Guide

## Part 1: Attach the Servo Motor to the Leg

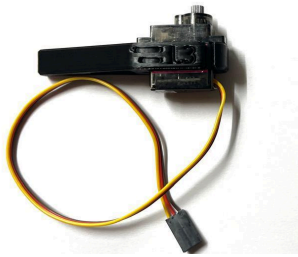
You are mounting the servo motor onto the leg so that when the servo rotates, it moves the tibia.

**Step 1.** Hold the tibia so the label faces you and is readable (the letter & number are not upside down).

**Step 2.** Hold the servo motor so the servo shaft faces up. The servo shaft is the silver, round, ridged, cylindrical piece at the top of the servo motor that rotates.

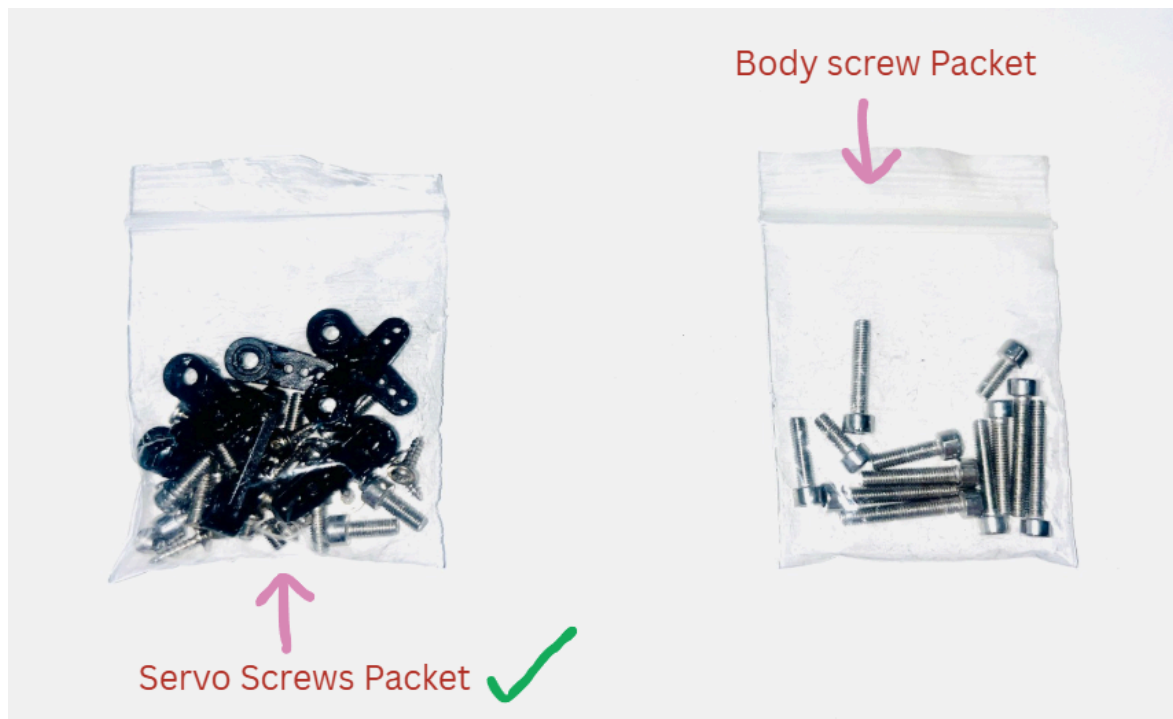


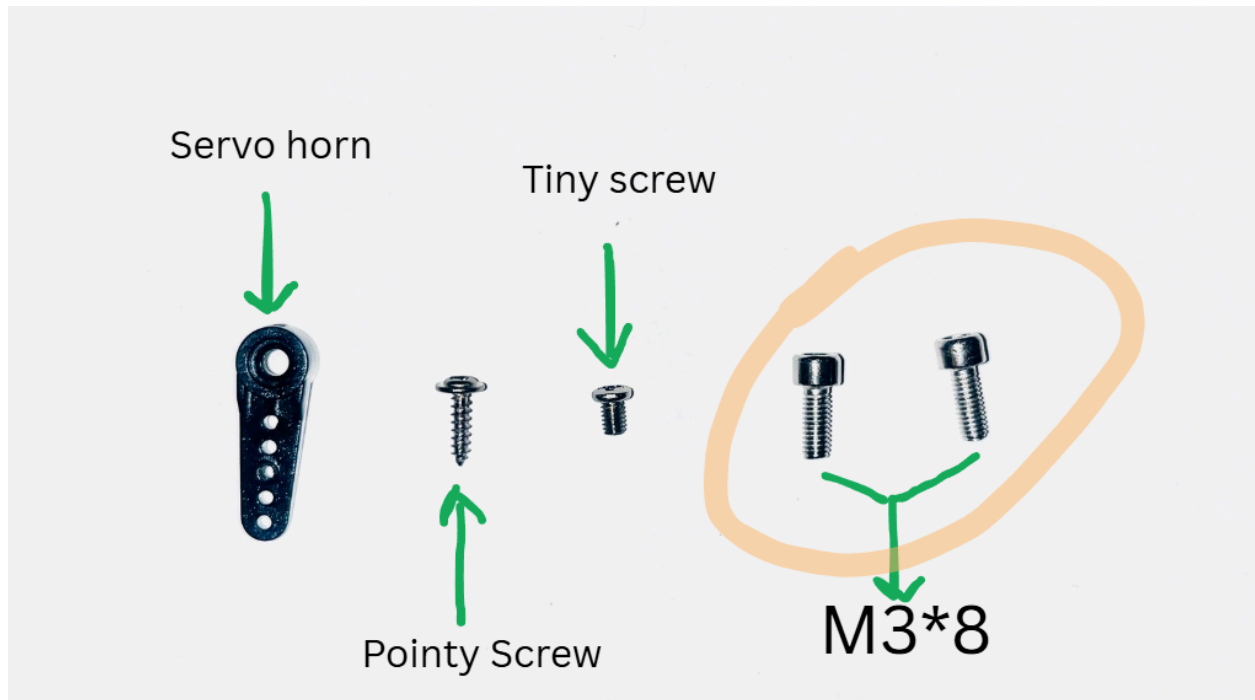
**Step 3.** Insert the servo motor into the rectangular cutout on the tibia, positioning it so the shaft is near the top of the leg and pointing upward, above the L3 label. The servo motor should sit flat against the tibia once it has been pushed in. Do not force it. If it does not sit flat, check the orientation and call an instructor.



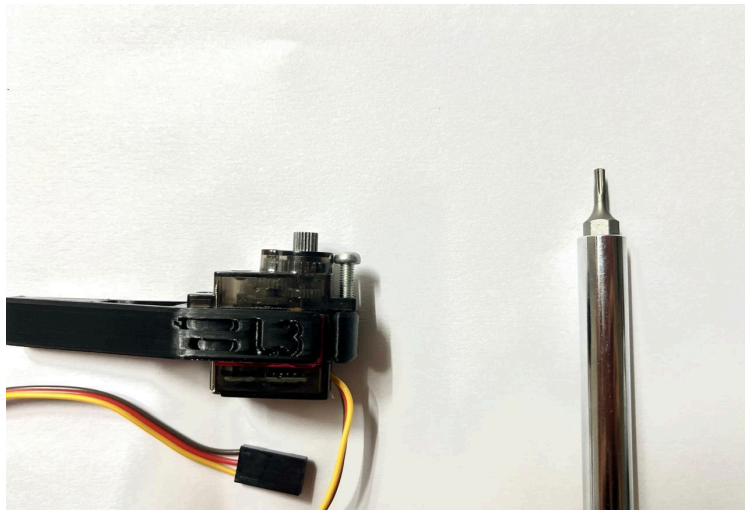
**Step 4.** Insert the H2.5 screwdriver tip into the screwdriver body. You can identify a tip as H2.5 by looking at the engraving on its side.

**Step 5.** You will have two packets. Use the servo screws packet. Take two 3x8 mm screws (hex screwhead)



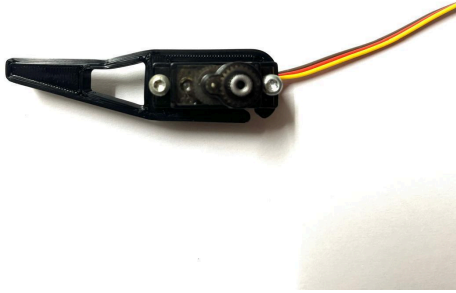


**Step 6.** Place one screw into the screw hole on one side of the servo motor and the tibia together. Turn the screwdriver right-ward/clockwise (tighten) slowly until the screwhead touches the servo. Do not overtighten as it may damage your motor and/or tibia.



**Step 7.** Repeat with the second screw on the other side of the motor.

**Step 8.** The servo motor is now firmly attached to the tibia. The top view of how the tibia should look is given below. Set the leg aside.



### **Q: Why Do We Set the Servo to 90 Degrees Before Attaching the Horn?**

A servo motor can rotate from 0 degrees to 180 degrees. The center position is 90 degrees.

For our quadruped robot (a four-legged walking robot), each tibia needs to move equally in both directions from the center. If the servo horn is attached while the servo is at the wrong angle, the leg will not be centered and will be able to move further in one direction than the other. By setting the servo to exactly 90 degrees before attaching the horn, we ensure the horn is physically centered so the leg can move the same amount in both directions.

### **Part 2: Connecting the Servo Motor to the ESP32**

You will now connect the servo motor to the XIAO ESP32-C3 using three male-to-female jumper wires.





### What the pins mean:

- **GND** — Ground. This is the negative reference point for all electrical signals. Every electronic component in a circuit must share a common ground to work correctly.
- **VUSB** — This pin provides power drawn from your laptop through the USB/C to C cable. The servo motor needs this power to move.
- **Do** — This is a digital input/output pin on the ESP32. Do means Digital Pin 0. It sends the signal to the servo motor telling it what angle to move to. Your ESP32 has many pins labelled Do, D1, D2, and so on, each one can send or receive signals independently.

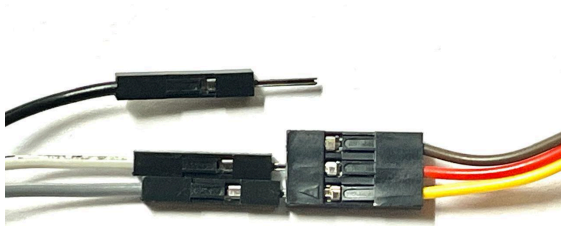


**Step 1.** Look at the servo motor. It has three wires coming out of it:

- Brown or black wire — this is Ground
- Orange/Red wire — this is Power
- Yellow wire — this is Signal

**Step 2.** Take one male-to-female jumper wire. Plug the female socket end onto the brown or black wire of the servo. Plug the male pin end into the GND pin on the ESP32.

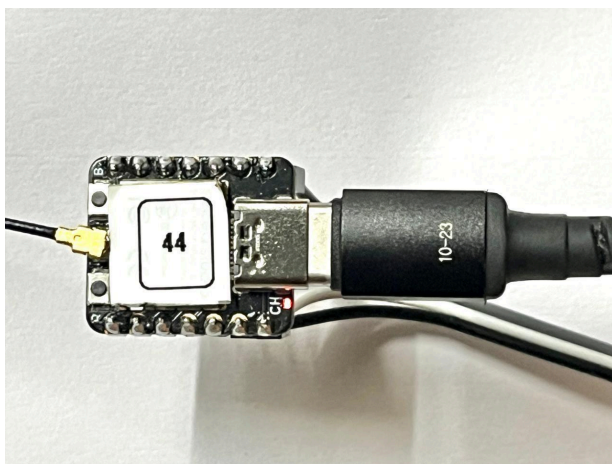
**Step 3.** Take a second male-to-female jumper wire. Plug the female socket end onto the red/orange wire of the servo. Plug the male pin end into the VUSB pin on the ESP32.



**Step 4.** Take a third male-to-female jumper wire. Plug the female socket end onto the yellow wire of the servo. Plug the male pin end into the D-0 pin on the ESP32.

**Why three wires?** Every component needs power (+), ground (–), and a way to receive instructions (signal).

**Step 5.** Plug the USB-C end of your cable into the ESP32. The red light on the ESP32 should blink or turn on, confirming it has power.



Check that the baud rate is set to **115200**. If it is not, change it to 115200. The baud rate is how many communication symbols are sent & received each second.

**Step 8.** In the text input box at the bottom of the Serial Monitor panel, type `90` and press Enter. You will see the message `-> Moved to 90 degrees`. The servo is confirmed at center position. When the servo moves, you will see the shaft rotating and hear a noise.

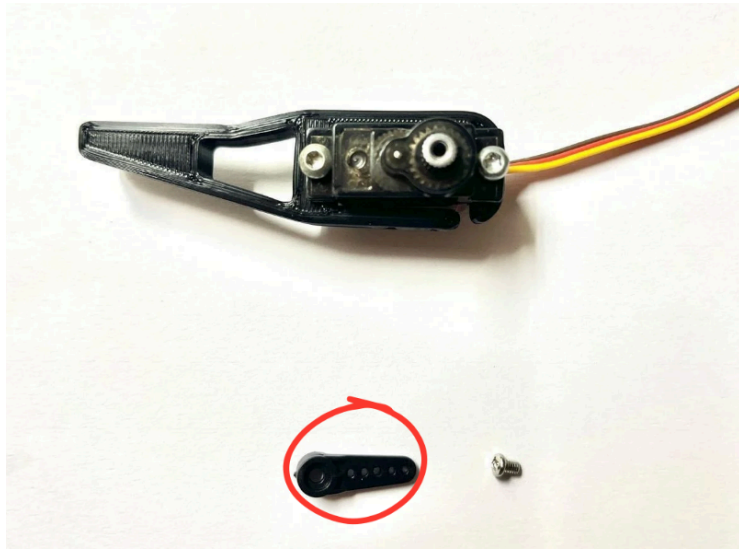
**Caution: If your servo is continuously making a noise for more than 5-10 seconds/getting heated (feels warm/hot when you touch the body), immediately disconnect the USB/C cable from your laptop and call an instructor.**

**Step 9.** You may now also type `0` and press Enter to see the servo move to one extreme, and `180` and press Enter to see it move to the other extreme (you can enter any angle for that matter!). Type `90` again to return to center before the next step.

**Step 10.** Unplug the USB cable from your laptop. The servo will hold its last position.

## **Part 6 Attach the Servo Horn**

**Step 1.** Take the servo horn from the servo horn packet. It is a plastic piece with an arm extending outward. It has been circled in the image below.

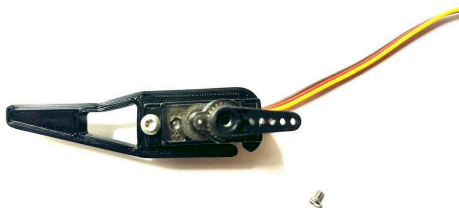


Use this for  
screwing  
servo horn on  
top of shaft



Look at the top of the servo motor. The output shaft has ridges around it. The servo horn has a matching ridged hole in its center. These ridges are designed to lock together so the horn cannot slip.

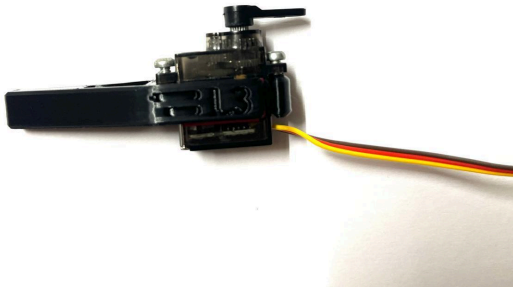
**Step 3.** Hold the tibia so the label faces you. Orient the servo horn so that one arm points straight outward away from the leg, at 180 degrees relative to the tibia body. This is the correct horn position (with respect to how it is going to be positioned in the main frame body).



**Step 4.** Press the horn down firmly onto the output shaft. You should feel it click into place. Do not rotate it once it is on, simply press straight down.

**Step 5.** Insert the PH01 screwdriver tip into the screwdriver body.

**Step 6.** Take the small servo horn screw from the servo horn packet. This screw is smaller than the 3x8mm screws, it is specifically sized for the center hole of the servo horn. It is not pointy.



**Step 7.** Place this screw into the hole of the servo horn. Turn the screwdriver clockwise until the screw is inside the cavity. Do not overtighten as this can crack the plastic horn.

---

## **Part 7 Attach the Hip Joint**

**Step 1.** Take the femur. Look at it carefully. It has two cavities (hollow openings) on it. One cavity is for the servo horn arm to fit into, and the other is for the body/frame. The cavity for the body frame has the label on its flip side. For the horn, use the other cavity. Slide the horn arm into this cavity. It should slide into this cavity perfectly and not be loose.



**Step 4.** Take the pointy screw from the servo horn packet. This screw secures the femur with the rest of the leg.

**Step 5.** Using the PH01 screwdriver tip, insert this screw through the hip joint and into the leg piece. Turn clockwise to tighten it.

## Part 8: Building the Remaining Four Legs

You have already built one leg. Now you will build four more legs using the exact same process. Each leg consists of a tibia, a servo motor, and a servo horn. Follow

the same steps you used in Parts 1, 2, and 6 for each new leg. Let's go over that again.

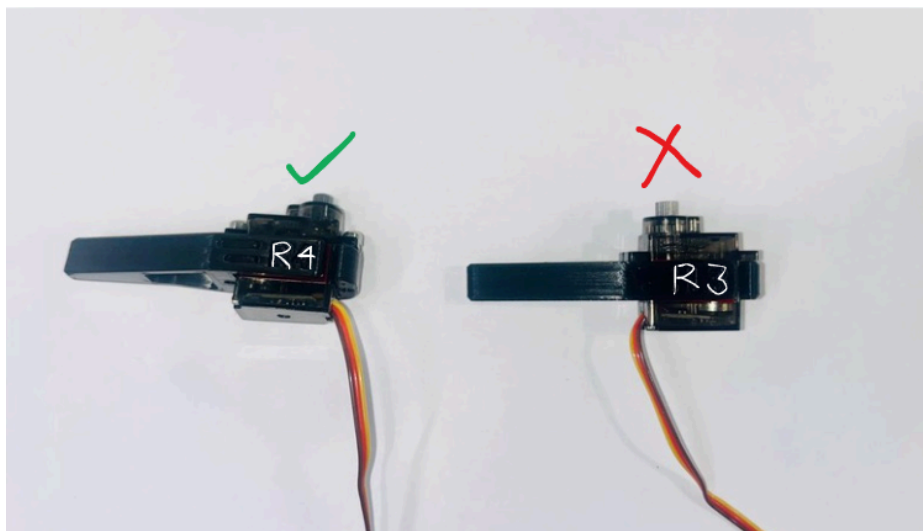


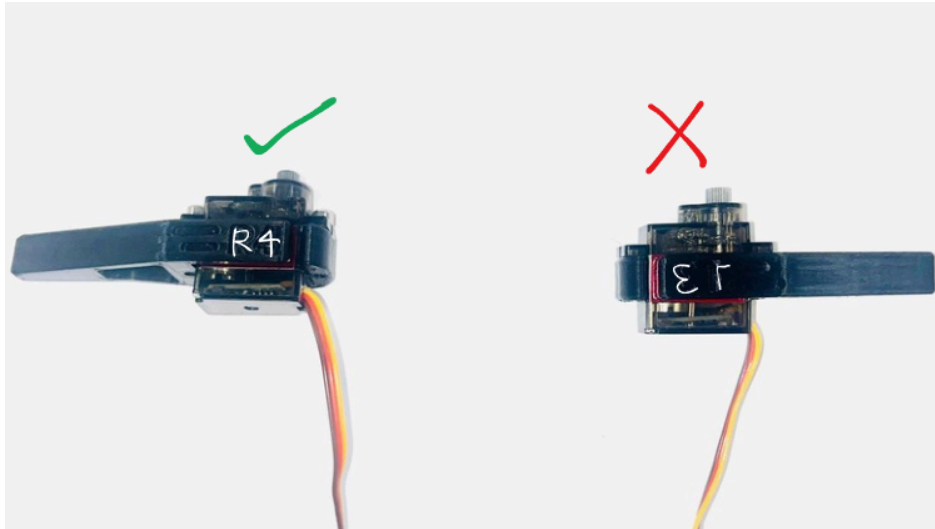
### **Step A Mount the Servo Motor to the Tibia**

**Step 1.** Hold the tibia so its label faces you and is readable.

**Step 2.** Hold the servo motor with the shaft (the silver, ridged, round cylinder) facing upward.

**Step 3.** Insert the servo motor into the rectangular cutout on the tibia, shaft near the top and pointing upward, above the leg label. The servo must sit flat, do not force it. If it does not sit flat, check orientation and call an instructor.





**Step 4.** Insert the H2.5 screwdriver tip into the screwdriver body.

**Step 5.** From the servo screws packet, take two 3×8 mm hex-head screws.

**Step 6.** Insert one screw into the hole on one side of the servo and tibia. Turn clockwise slowly until the screwhead is snug against the servo. Do not overtighten, this can damage the motor or tibia.

**Step 7.** Repeat with the second screw on the other side. The servo is now firmly attached to this tibia.





## **Step B Connect the Servo to the ESP32 and Set to 90 Degrees**

**Step 1.** Connect the three servo wires to the ESP32 using male-to-female jumper wires:

- Brown/black wire → GND pin on the ESP32
- Red/orange wire → VUSB pin on the ESP32
- Yellow wire → Do pin on the ESP32

**Step 2.** Plug the USB-C cable into the ESP32. The red light should blink or turn on.

**Step 3.** Open the Serial Monitor in your IDE and confirm the baud rate is set to 115200.

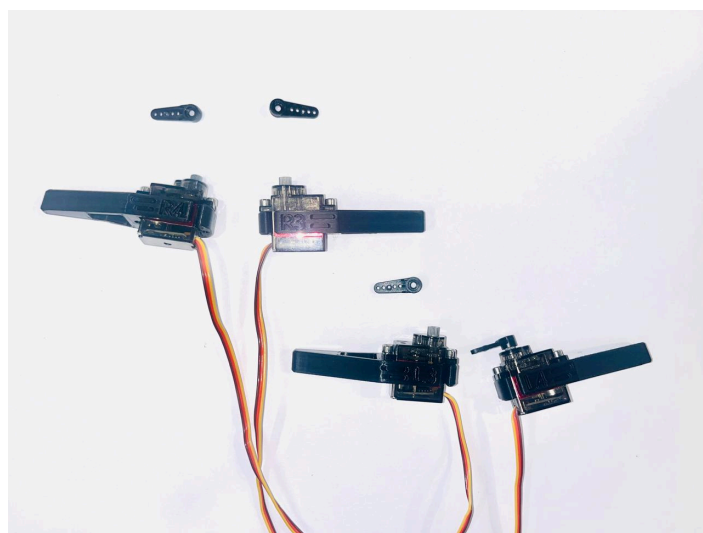
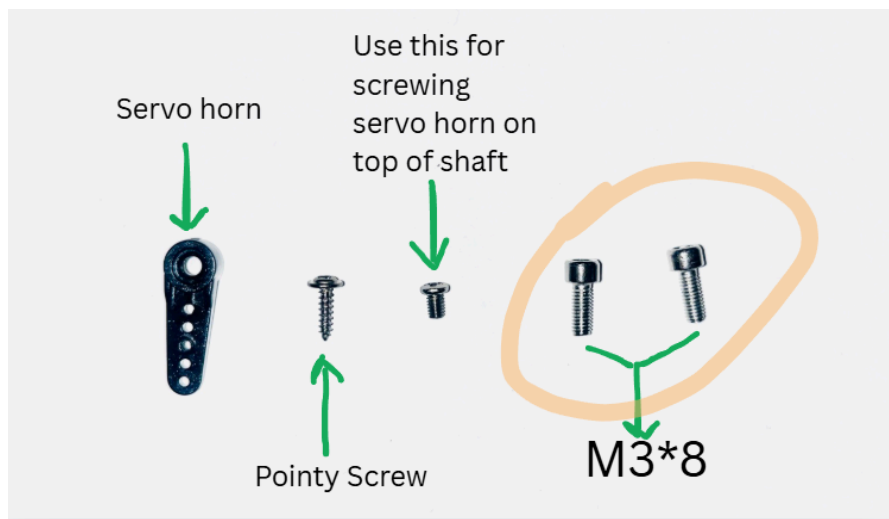
**Step 4.** Type "90" in the Serial Monitor input box and press Enter. You should see the message: Moved to 90 degrees. The servo shaft will rotate and you may hear it move.

 **CAUTION:** *If the servo makes a continuous noise for more than 5–10 seconds or feels warm/hot to the touch, immediately unplug the USB-C cable from your laptop and call an instructor.*

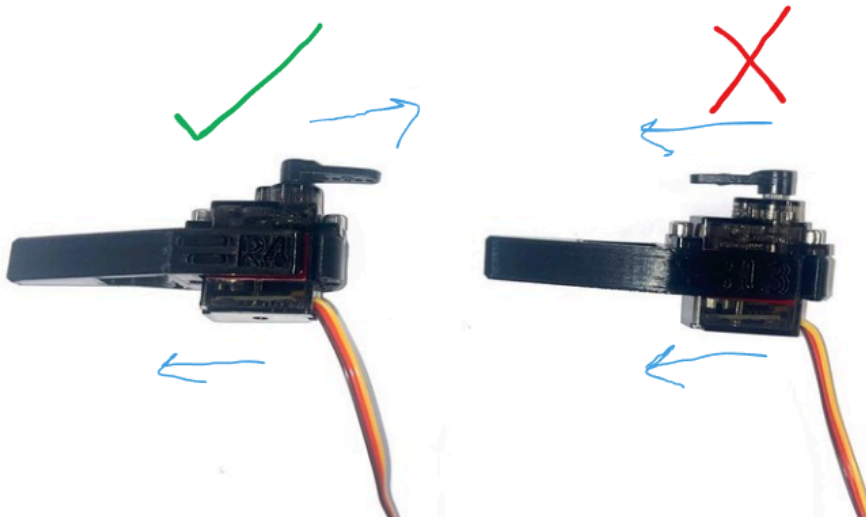
**Step 5.** Once confirmed at 90 degrees, unplug the USB-C cable. The servo will hold its position.

## **Step C Attach the Servo Horn**

**Step 1.** Take a servo horn from the servo horn packet. It is a plastic piece with an arm extending from a central hub with a ridged hole.



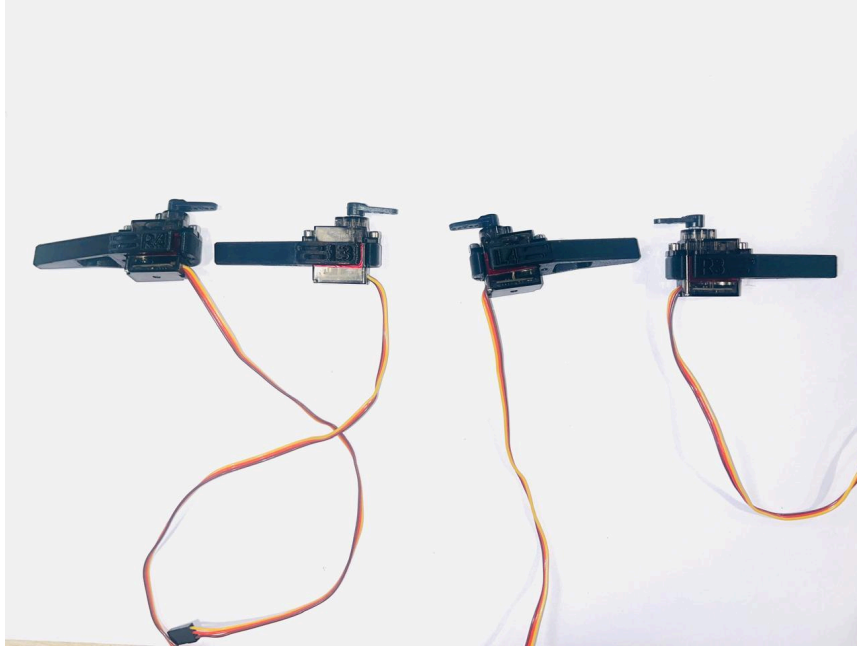
**Step 2.** Hold the tibia so its label faces you. Orient the servo horn so that the arm points straight outward away from the leg body, this is the 180-degree position relative to the tibia. This is the correct orientation for how the leg will sit in the main frame.



**Step 3.** Press the horn firmly and straight down onto the servo shaft. You should feel it click. Do not rotate it once it is on the shaft; press straight down only.

**Step 4.** Insert the PH01 screwdriver tip into the screwdriver body.

**Step 5.** Take the small servo horn screw from the servo horn packet (it is smaller than the 3×8 mm screws and is not pointed). Place it into the center hole of the horn. Turn clockwise until snug. Do not overtighten; this can crack the plastic horn.



## Leg Labeling and Pairing

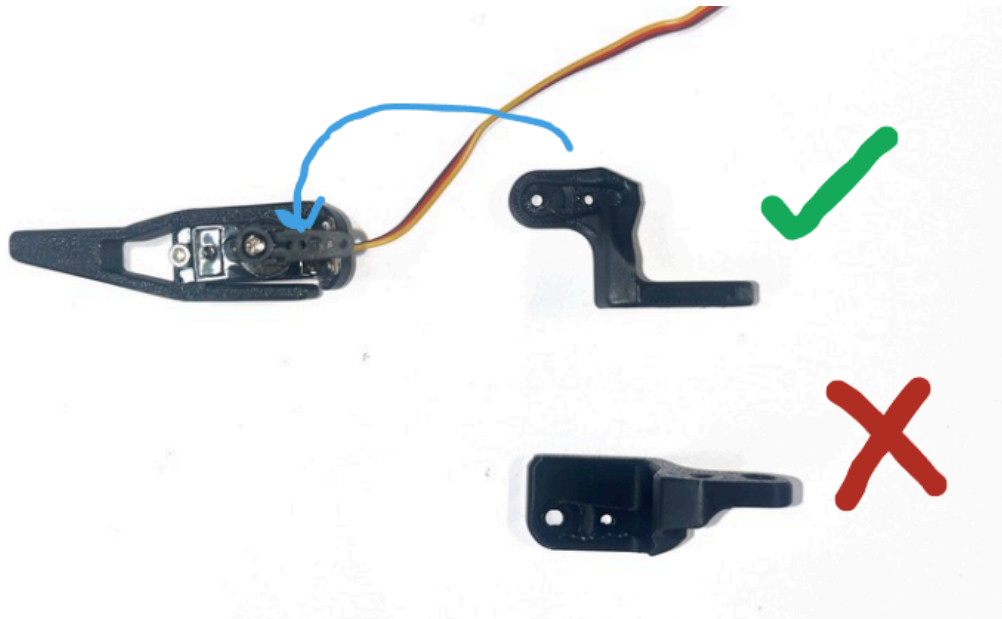
Your robot has four legs. Each leg is identified by a side (L for Left, R for Right) and a number. The pairings that will be assembled onto the same side of the body frame are:

- R1 and R4 Right side, front and rear
- L1 and L4 Left side, front and rear
- R2 and R3 Right side, inner front and inner rear
- L2 and L3 Left side, inner front and inner rear

Keep paired legs together as you build them so they are easy to identify during final assembly.

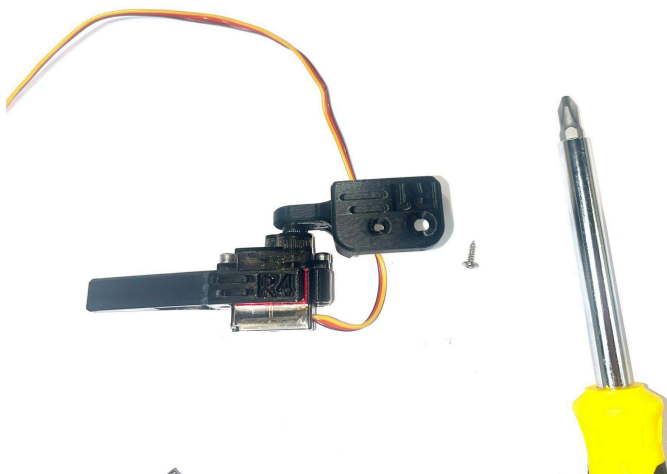
### **Step D Attach the Hip Joint**

**Step 1.** Take the femur piece. Examine it carefully, it has two cavities. The cavity whose reverse side has a label is for the body frame. The other cavity is for the servo horn arm.

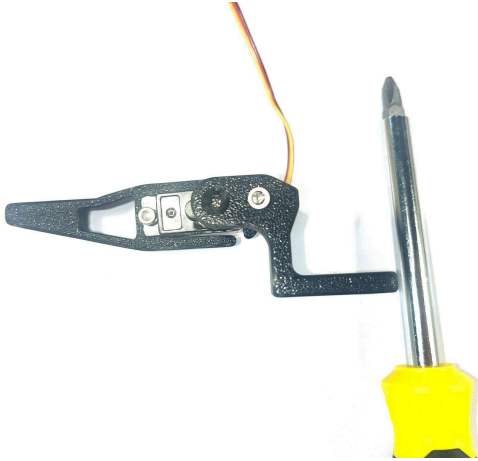


**Step 2.** Slide the servo horn arm into the non-labeled cavity. It should fit snugly without any looseness.

**Step 3.** Take the pointed screw from the servo horn packet.



**Step 4.** Using the PH01 screwdriver tip, insert the pointed screw through the hip joint and into the leg piece. Turn clockwise to tighten and secure the femur.



Repeat Steps A through D for all four remaining legs. Before moving on, verify:

- All four legs have their servo mounted and screwed into the tibia.
- All four servos were set to 90 degrees before the horn was attached.
- All four servo horns are attached with the arm pointing outward (180 degrees relative to the tibia body).
- All four hip joints (femurs) are secured with the pointed screw.

### **Why Do We Set the Servo to 90 Degrees Before Attaching the Horn?**

A servo motor rotates from 0° to 180°, with 90° being the exact center. For the robot to walk properly, each leg must be able to swing equally in both directions. If the servo horn is attached while the servo is off-center, the leg will reach further in one direction than the other. Setting the servo to 90° first ensures the horn is physically centered, giving equal range of motion in both directions.

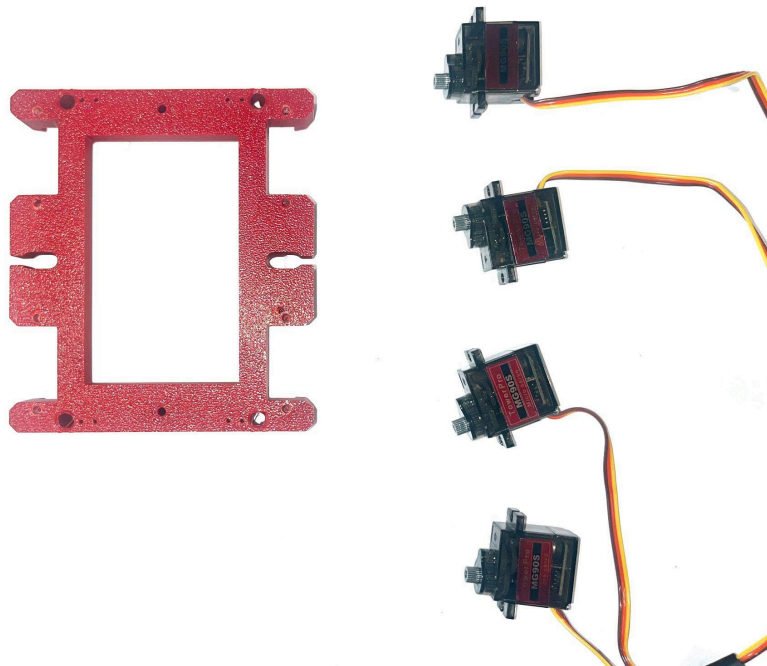
## **Part 9: Attaching the Body Servos to the Main Frame**

You will now mount four servo motors directly onto the main body frame. These are the hip servos, they rotate the entire leg assembly by moving the femur. This is different from the servos already inside your legs (which move the tibia). The hip servos sit in the body and connect to the leg through the hip joint you just built.

## Orientation Before You Begin

Place the main body frame on your work surface with:

- The wiring channels (slots where wires will run) facing toward the table — facing downward.
- The top surface of the body facing upward.



## Step 1 Set All Four Body Servos to 90 Degrees

Before mounting any servo to the frame, you must set each one to 90 degrees. This is because the horn will be attached at a diagonal (45 degrees) relative to the frame, and setting it to 0 first establishes a known physical reference point.

**Step 1.** Connect a servo motor to the ESP32 using three male-to-female jumper wires:

- Brown/black wire → GND
- Red/orange wire → VUSB

- Yellow wire → Do

**Step 2.** Plug in the USB-C cable. Open the Serial Monitor (baud rate 115200).

**Step 3.** Type "help" and then "90" in the Serial Monitor and press Enter. The servo will move to 90 degrees.

**Step 4.** Unplug the USB-C cable. The servo holds its position at 0 degrees.

**Step 5.** Repeat for all four body servos.

### Why Do We Set the Body Servos to 90 Degrees?

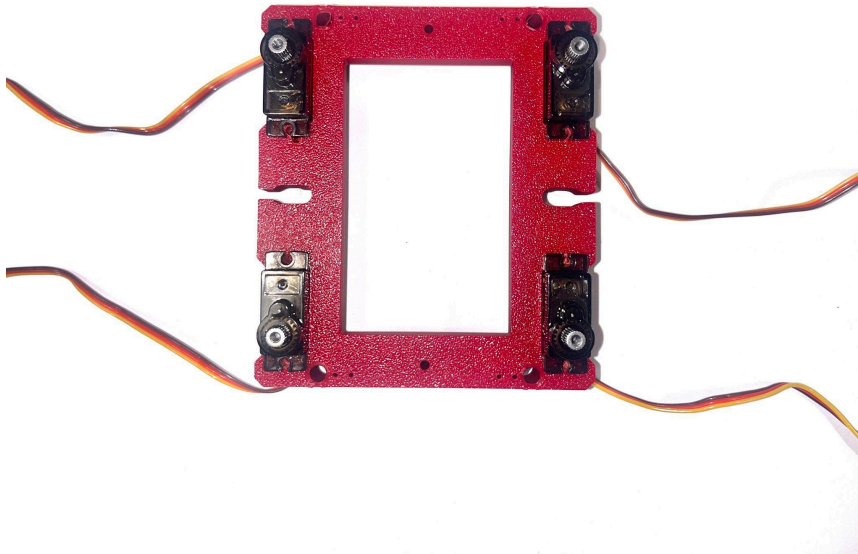
The body frame has specific geometry. The servo horn will be attached at 45 degrees (or 135 degrees) diagonally relative to the frame edges. To achieve this known angle precisely, we start with the servo at 90-degree and then orient the horn at 45 degrees relative to the frame. This ensures that when the servo sweeps its range, the leg sweeps symmetrically and in the correct plane for walking.

## Step 2 Mount the Servos to the Body Frame

**Step 1.** Take one servo motor (set to 90 degrees).

**Step 2.** Position the servo in its designated mounting location on the body frame. The servo shaft must face outward pointing away from the center of the body, toward where the leg will attach.





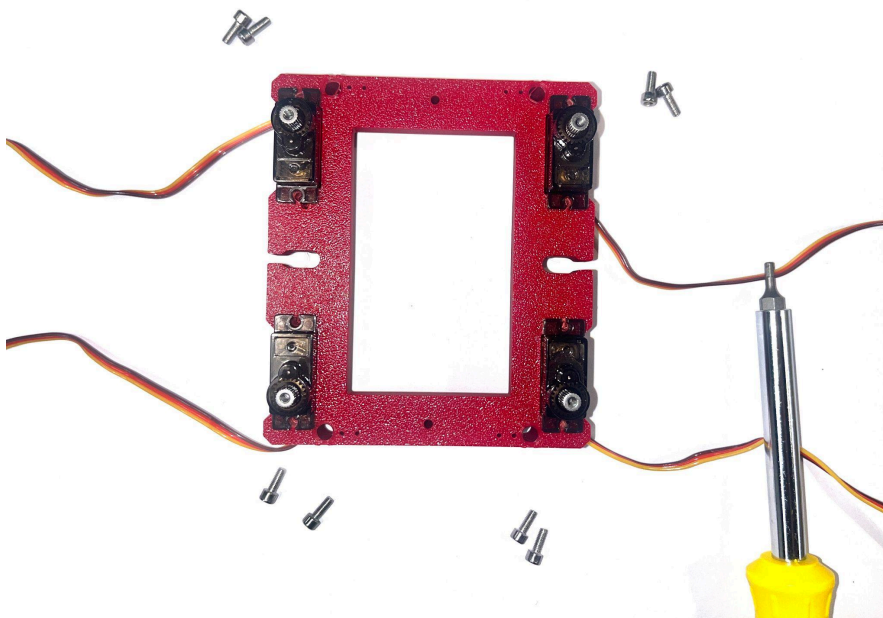
**Step 3.** Insert the H2.5 screwdriver tip into the screwdriver body.

**Step 4.** From the servo screws packet, take two 3×8 mm hex-head screws.

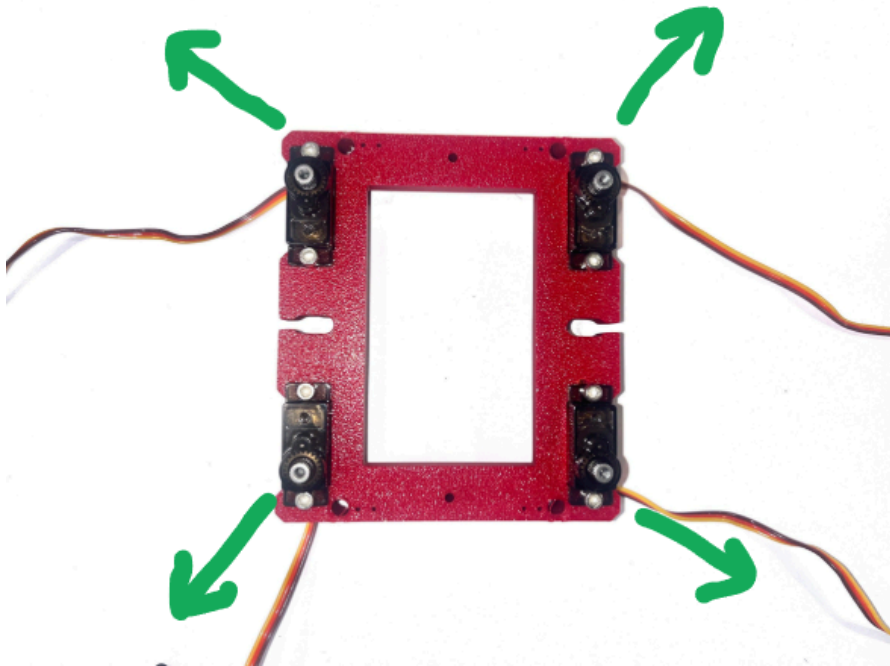
**Step 5.** Insert one screw into the mounting hole on one side of the servo and body frame together. Turn clockwise until snug. Do not overtighten.

**Step 6.** Insert the second screw into the mounting hole on the other side. Tighten.

**Step 7.** Repeat for all four servos, one at each leg position.



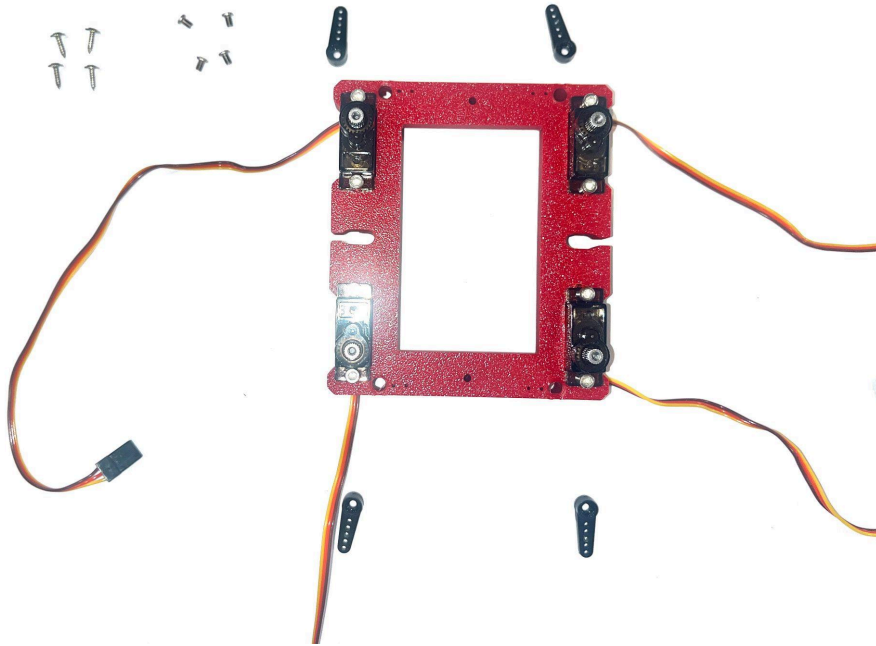
**⚠ NOTE:** *Verify that all four servo shafts are facing outward from the body frame before continuing.*



### **Step 3 Attach the Horn to Each Body Servo at 45 Degrees**

With the body servos mounted and sitting at 0 degrees, you will now attach the servo horn. The horn must be angled at 45 degrees (or 135 degrees for the opposite side) relative to the frame edges diagonally. This is the resting position that allows the legs to swing both forward and backward symmetrically during walking.

**Step 1.** Take a servo horn from the servo horn packet.

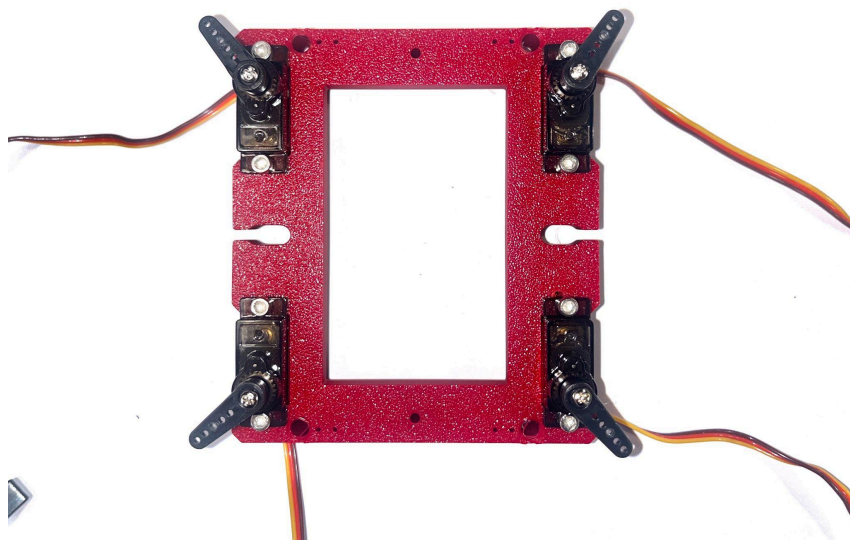


**Step 2.** Hold the body frame so you can see the servo shaft clearly. Orient the servo horn so its arm points diagonally at 45 degrees to the frame edge for the front servos, and 135 degrees for the rear servos, as appropriate. The exact diagonal direction ensures the leg hangs down naturally when assembled.

**Step 3.** Press the horn straight down firmly onto the shaft. You should feel it click into place. Do not rotate it, press straight down.

**Step 4.** Insert the PH01 screwdriver tip.

**Step 5.** Take the small servo horn screw from the servo horn packet. Place it in the horn's center hole. Turn clockwise until snug. Do not overtighten.



**Step 6.** Repeat for all four body servo horns.

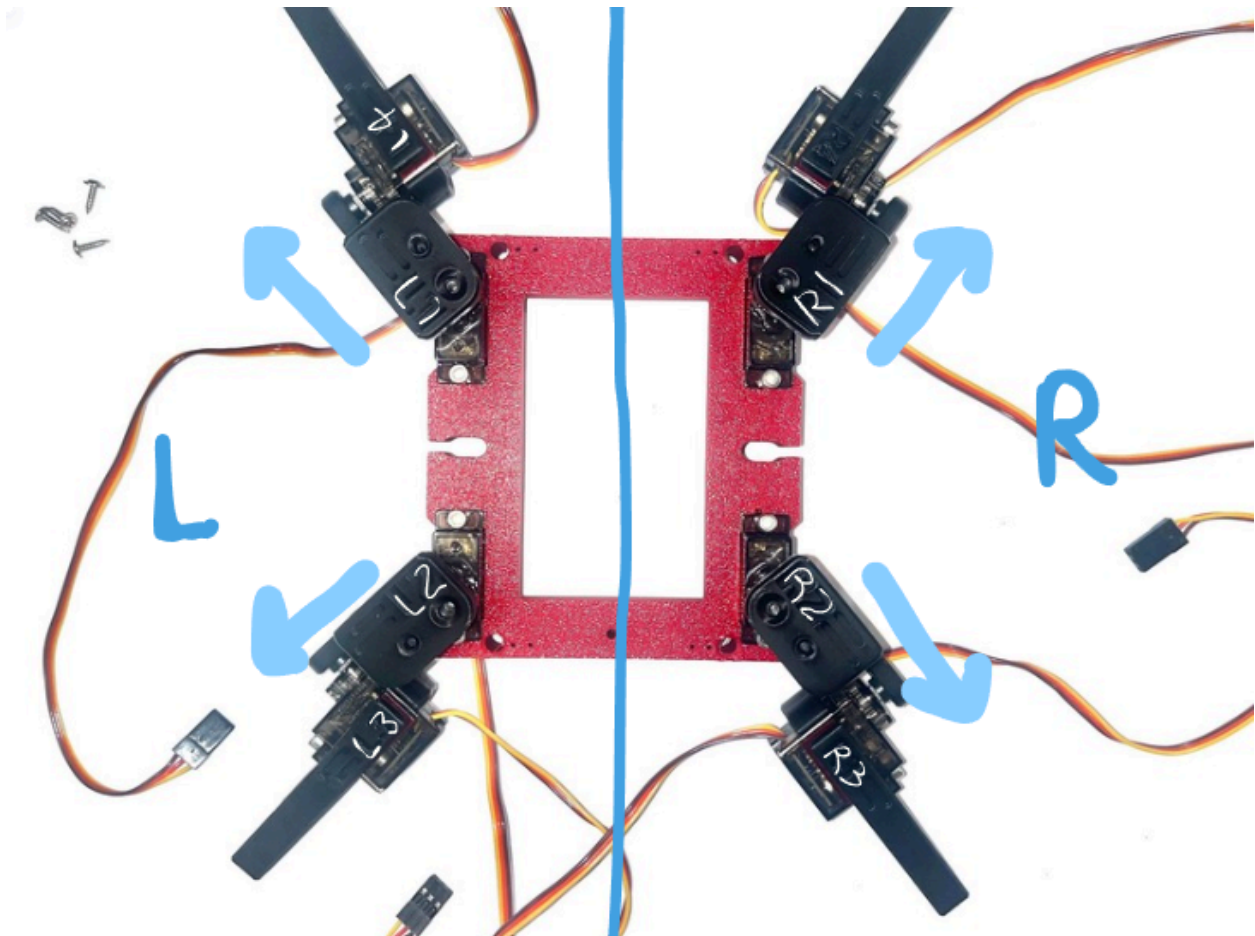
## Part 10: Attaching the Leg Assemblies to the Body

You will now connect each finished leg assembly to the body frame by attaching the assembly of femur, servo and tibia onto the servo horn you just mounted on the body.

## Leg Assignment

Attach legs to their correct positions on the body frame according to the pairings below. Each pairing goes on the same side of the body:

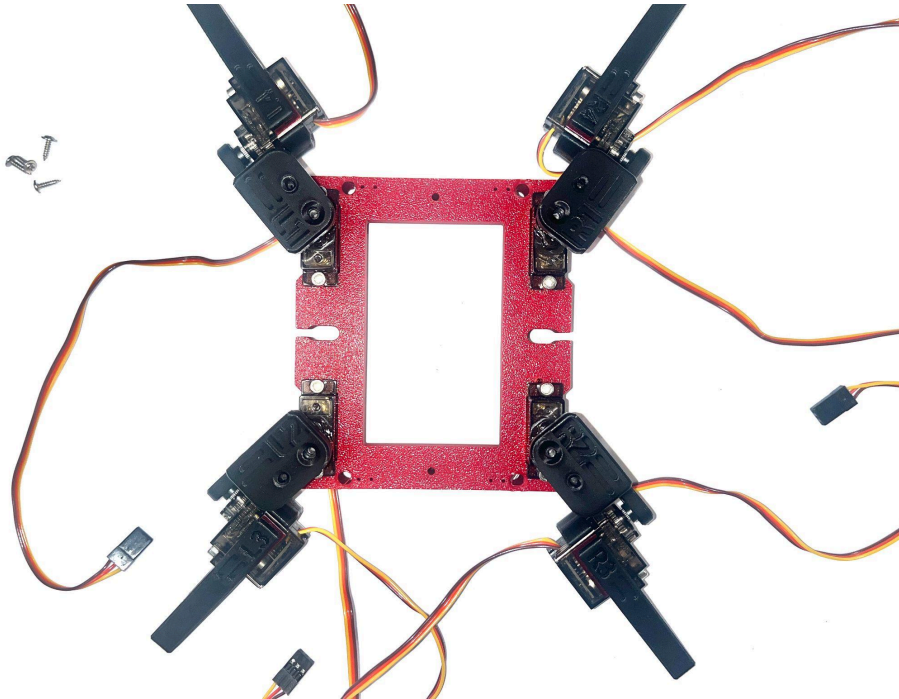
- R1 and R4 together — right side of the body
- L1 and L4 together — left side of the body
- R2 and R3 together — right side, inner positions
- L2 and L3 together — left side, inner positions



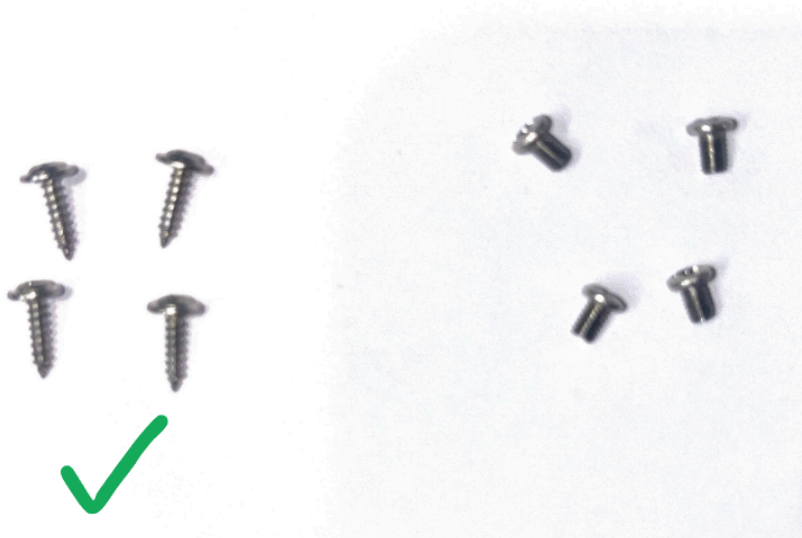
## Steps For Each Leg

**Step 1.** Take the leg assembly for the correct position. The hip joint (femur) has a cavity whose reverse side is labeled, this cavity mates with the body frame side. The label-side cavity faces the body.

**Step 2.** Slide the horn arm (now attached to the body servo) into the labeled cavity of the femur. It should fit snugly.



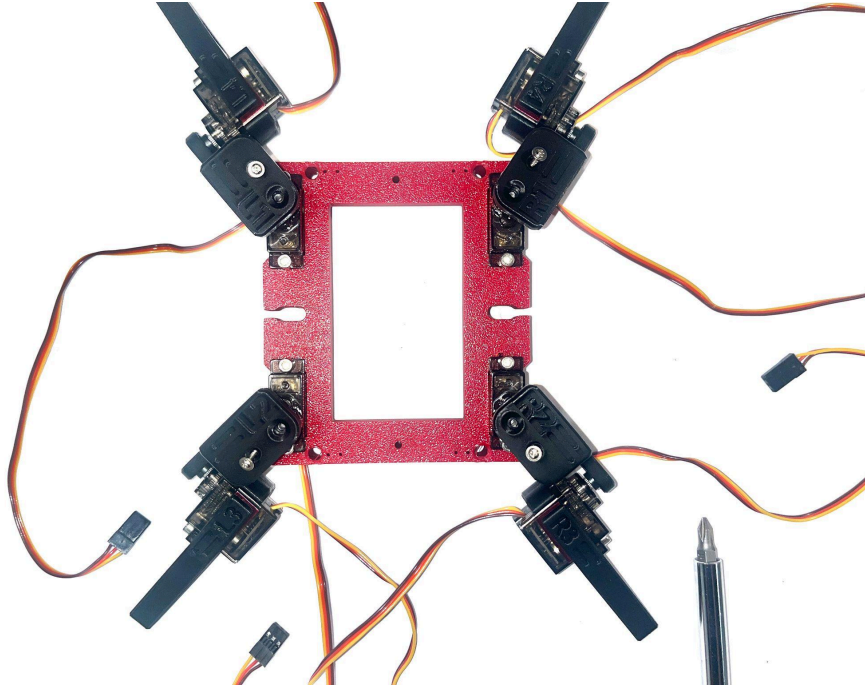
**Step 3.** Take the pointed screw from the servo horn packet.





**Step 4.** Using the PH01 screwdriver tip, insert the pointed screw through the hip joint into the leg piece from the frame side. Turn clockwise to secure.

**Step 5.** Repeat for all four leg assemblies.



## Part 11: Making the Robot Stand

With all four legs attached, you will now connect all four servos to the ESP32 and command them to find the standing position. This is the first time the robot will stand on its own four legs.

### Step 1 Connect Four Leg Servos to the ESP32 one at a time

The leg servo needs its three wires connected to the ESP32.

Connect wires to: GND, VUSB, and Do.

## Step 2 Set Legs to Standing Position

The standing position is achieved by commanding the leg servos to 0 degrees or 180 degrees whichever results in the legs extending downward and the body lifting off the table. Try both:

**Step 1.** Type "0" in the Serial Monitor and press Enter. Observe whether the leg extends outward/downward into a standing posture.

**Step 2.** If the leg does not stand at 0 degrees, type "180" and press Enter to try the other extreme.

**Step 3.** Once the standing position is found, note whether it is 0 or 180 degrees this will be used as the baseline in your walking code.

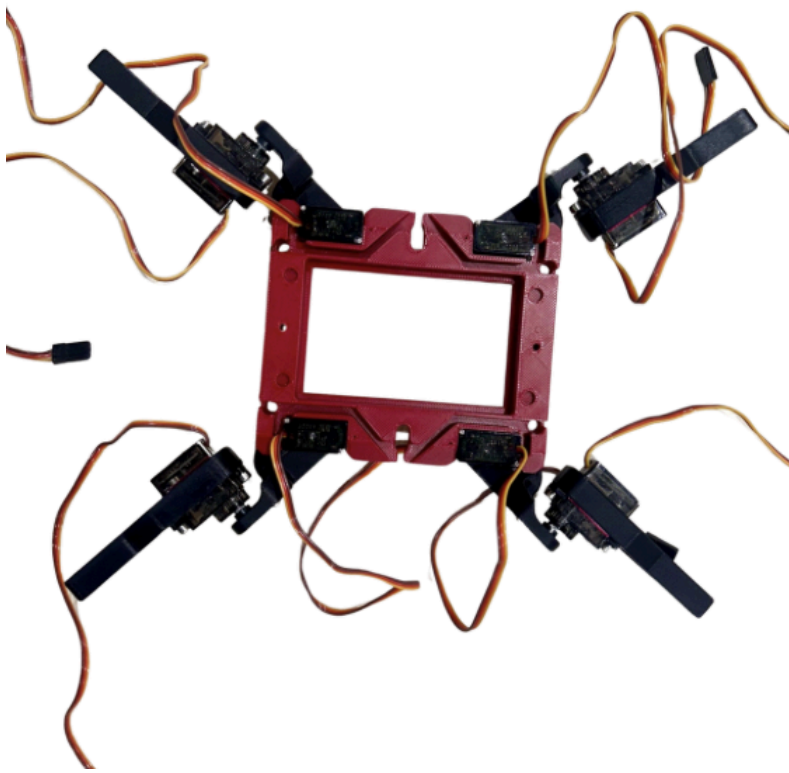
 **CAUTION:** *If a servo makes a continuous noise for more than 5–10 seconds or feels warm/hot, immediately unplug the USB-C cable from your laptop and call an instructor.*



## PART 12: Wire Management

Now, with all legs and hips attached along with their motors, you will slot the wires into the frame and place them into their respective pins on the Servo Driver.

Additionally, you will attach the bottom plate, along with the PCB and the Servo Driver, to the frame.



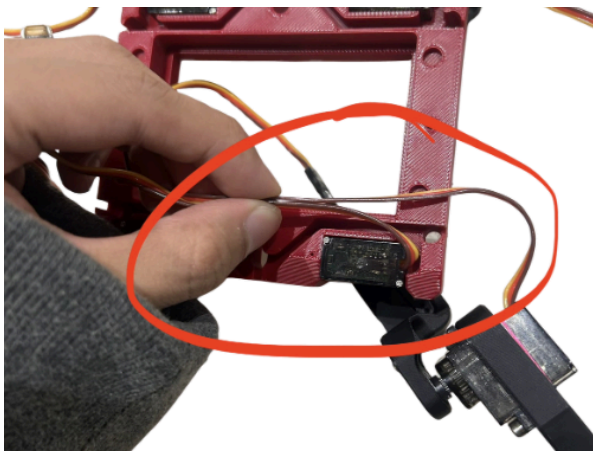
### Step 1: Slot the wires into the frame of the body

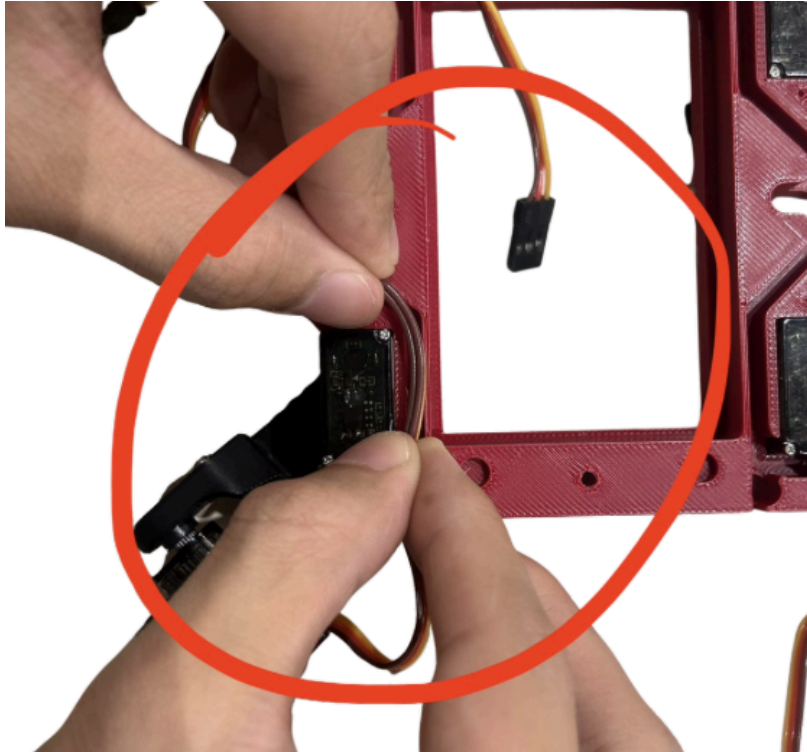
**Step 1:** Take the wires of one of the leg motors and put them on the groove on the frame.



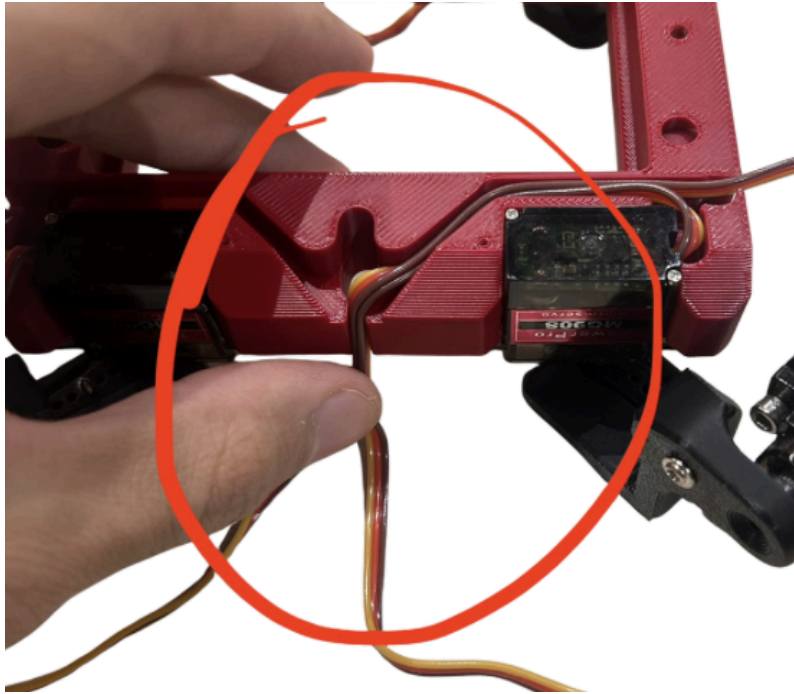
**Important:** Remember to leave some of the wire hanging out to allow for movement. A tightly slotted wire will constrict the movement of the motors. Also your **WIRES WILL SNAP AND RONNIE WILL NOT MOVE!**

**Step 2:** Now take the wire of the corresponding hip motor and place them together before slotting both together into the groove present on the frame.

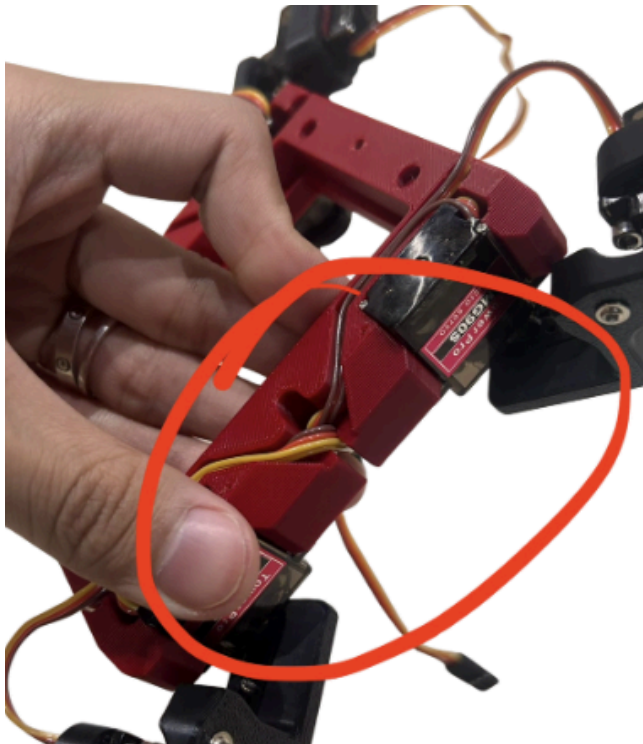




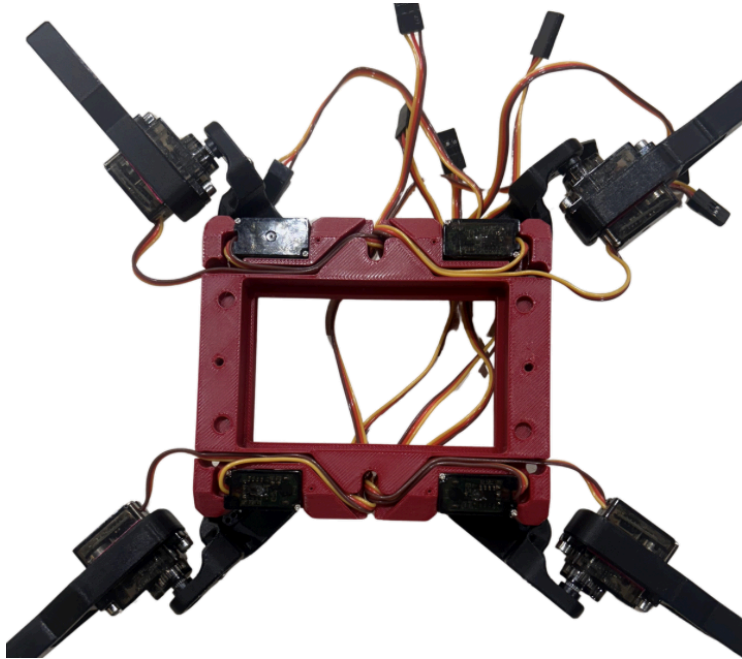
**Step 3:** After slotting both pairs of wires into the grooves, insert them into the hole as depicted in the picture below.



**Step 4:** Repeat for all remaining legs. Ensure that the wires on any one side of the frame appear as depicted in the image.



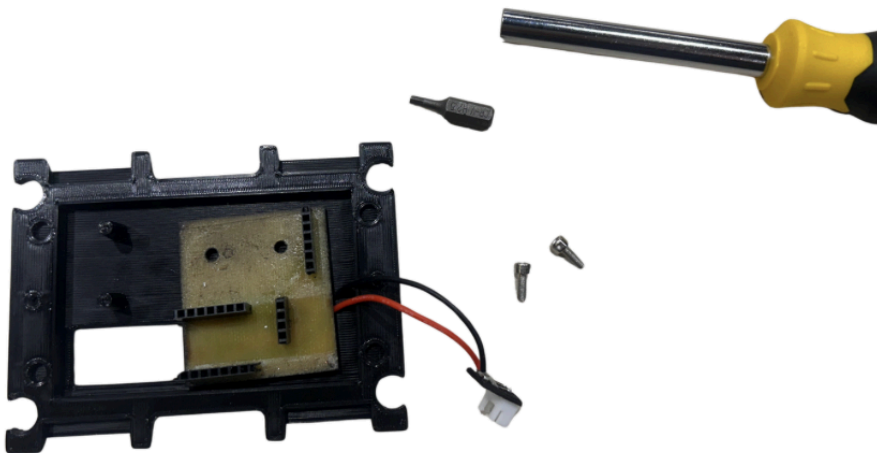
Once done, your wires should look like so



**Step 5:** Take the bottom plate present in the kit along with the PCB and two M3x4 screws, along with the Hex 2.5 screw head. Attach the PCB to the bottom plate. Ensure the wire is present on the PCB, faces down with the Pin Head slots facing up.

**Important:** The wire on the PCB must face outwards as depicted in the picture.

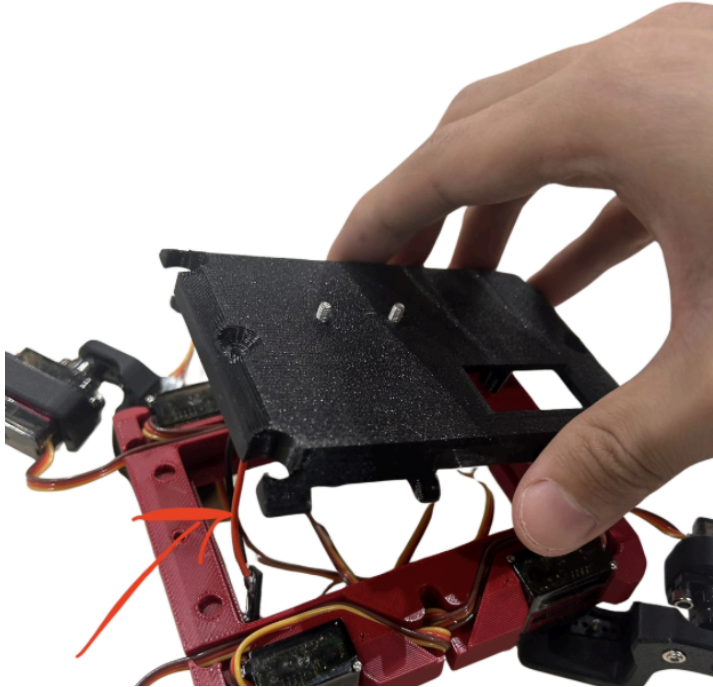
**Note:** You will have to apply a little force while screwing this bit.



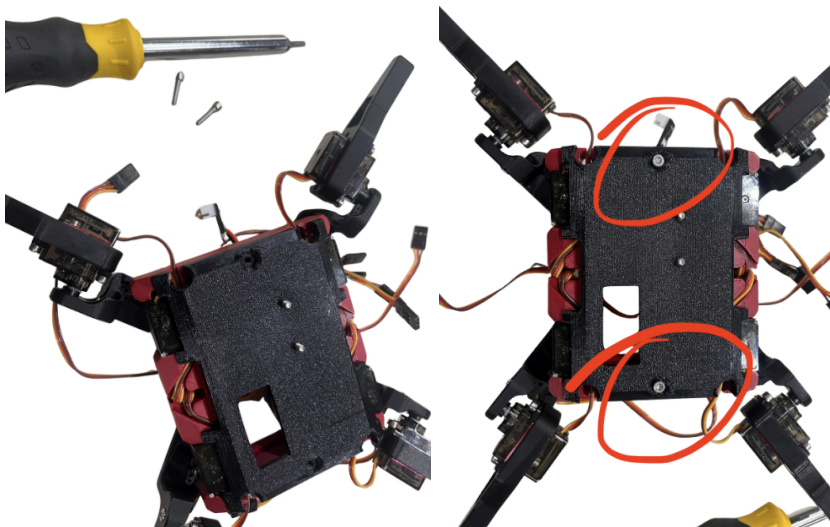
**Step 6:** Place the bottom plate onto the frame so that the PCB wire fits into the cavity.

**PLEASE ENSURE YOU DO NOT WEDGE THE WIRE IN BETWEEN THE BOTTOM PLATE AND THE FRAME!**

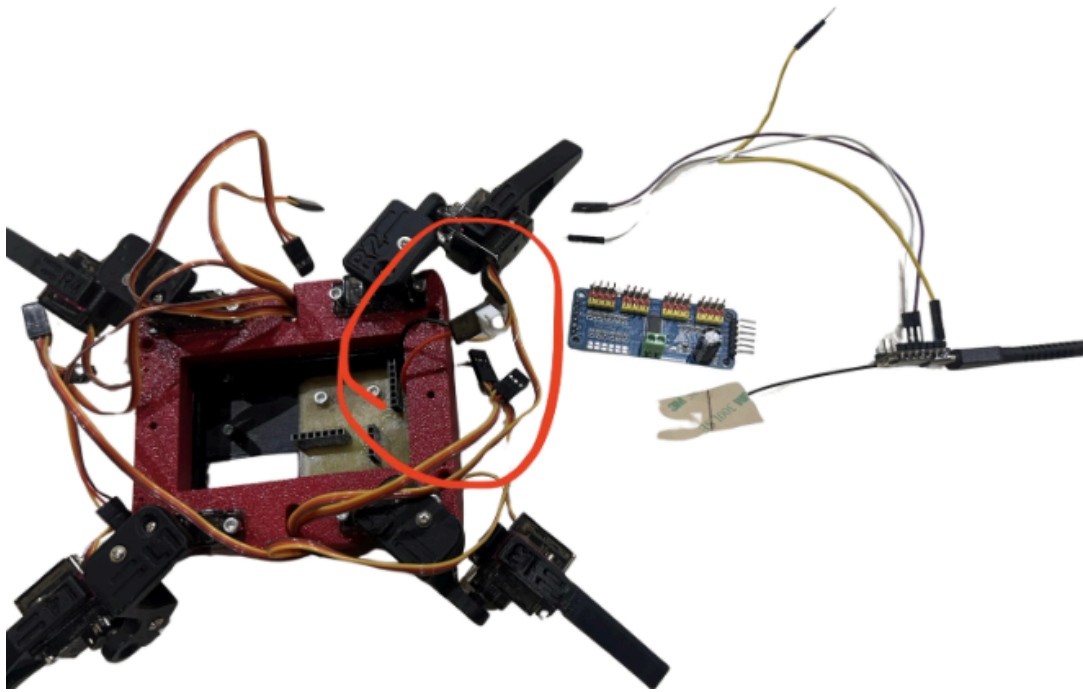




**Step 7:** Take two M3x12 screws from the second screw packet provided in the kit and screw the bottom plate onto the frame.



**Step 8:** Now, take the Servo Driver out of the kit along with your ESP 32.



**Step 9:** Test each motor now and attach it to its corresponding pin slot according to the diagram below.

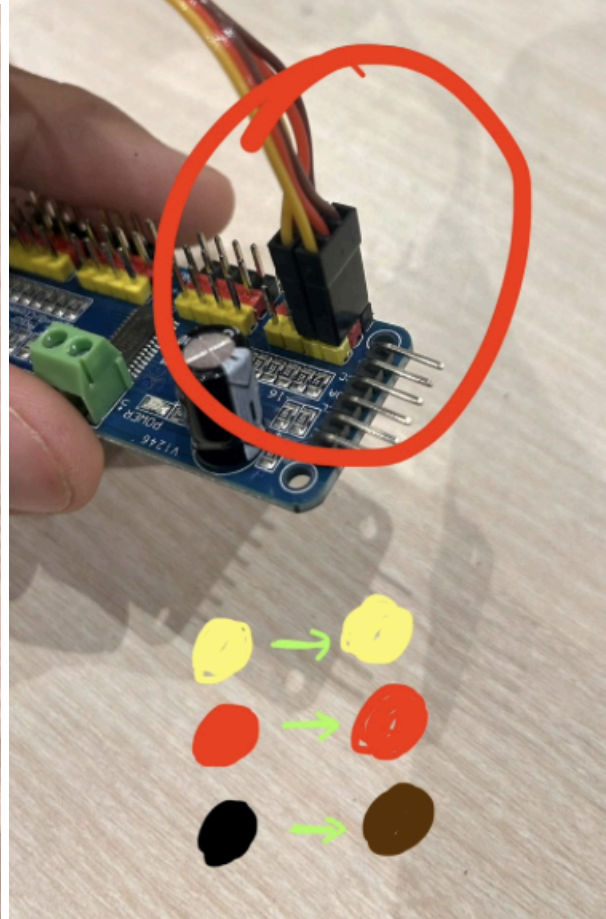
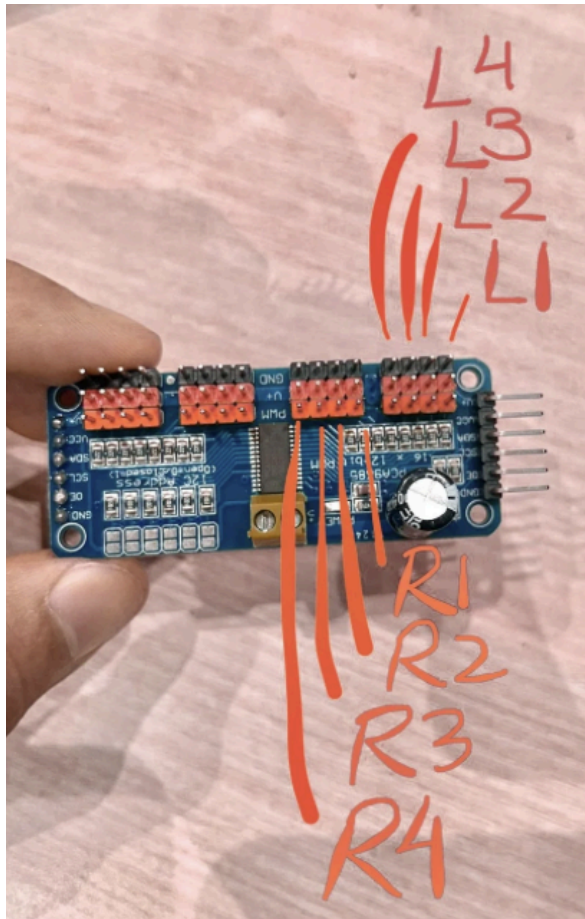
**Please ensure to maintain this order.**

Note that the colors on the motor's wires correspond to the colors present on the Servo Driver.

**Yellow to Yellow**

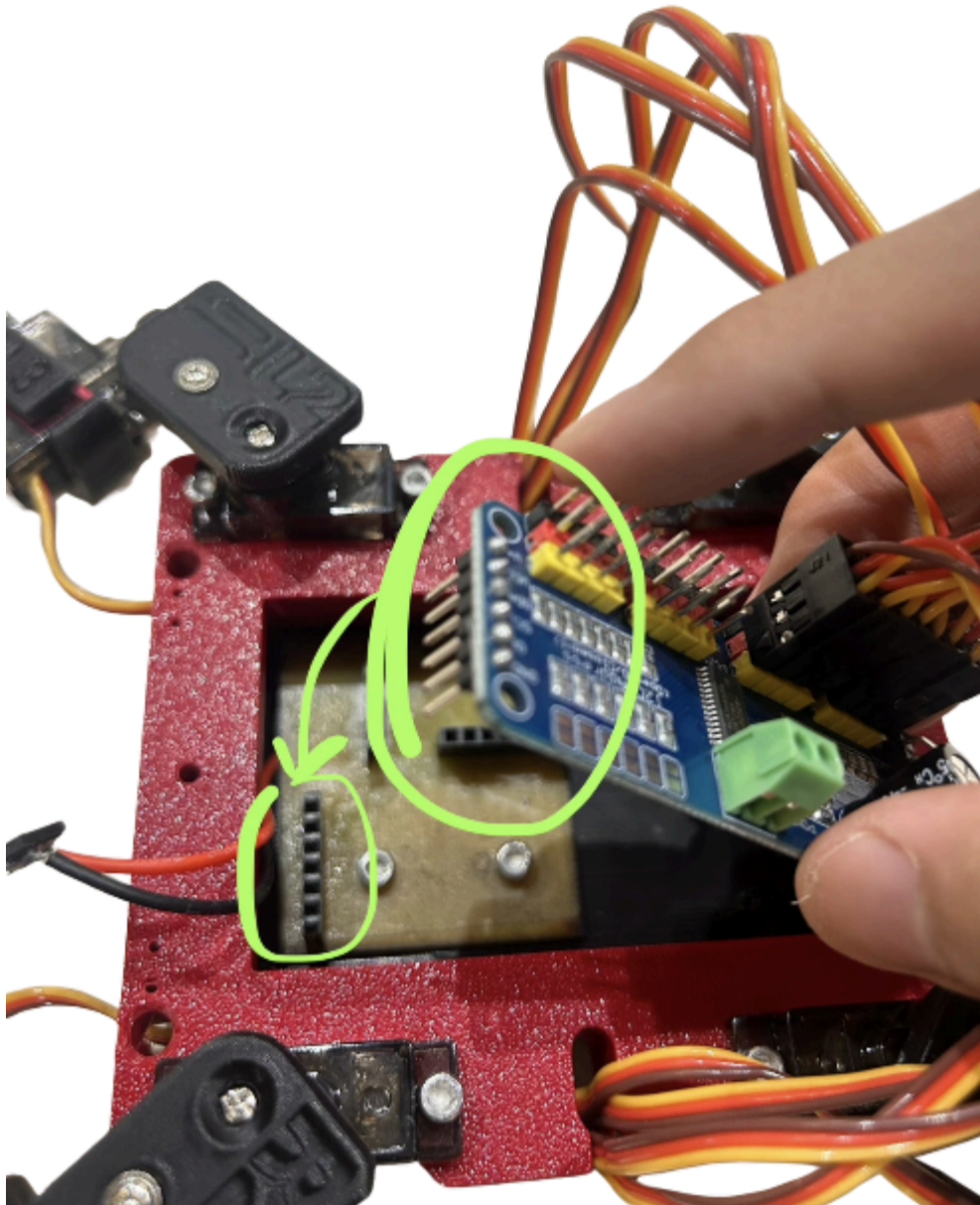
**Red to Red**

**Black to Brown**



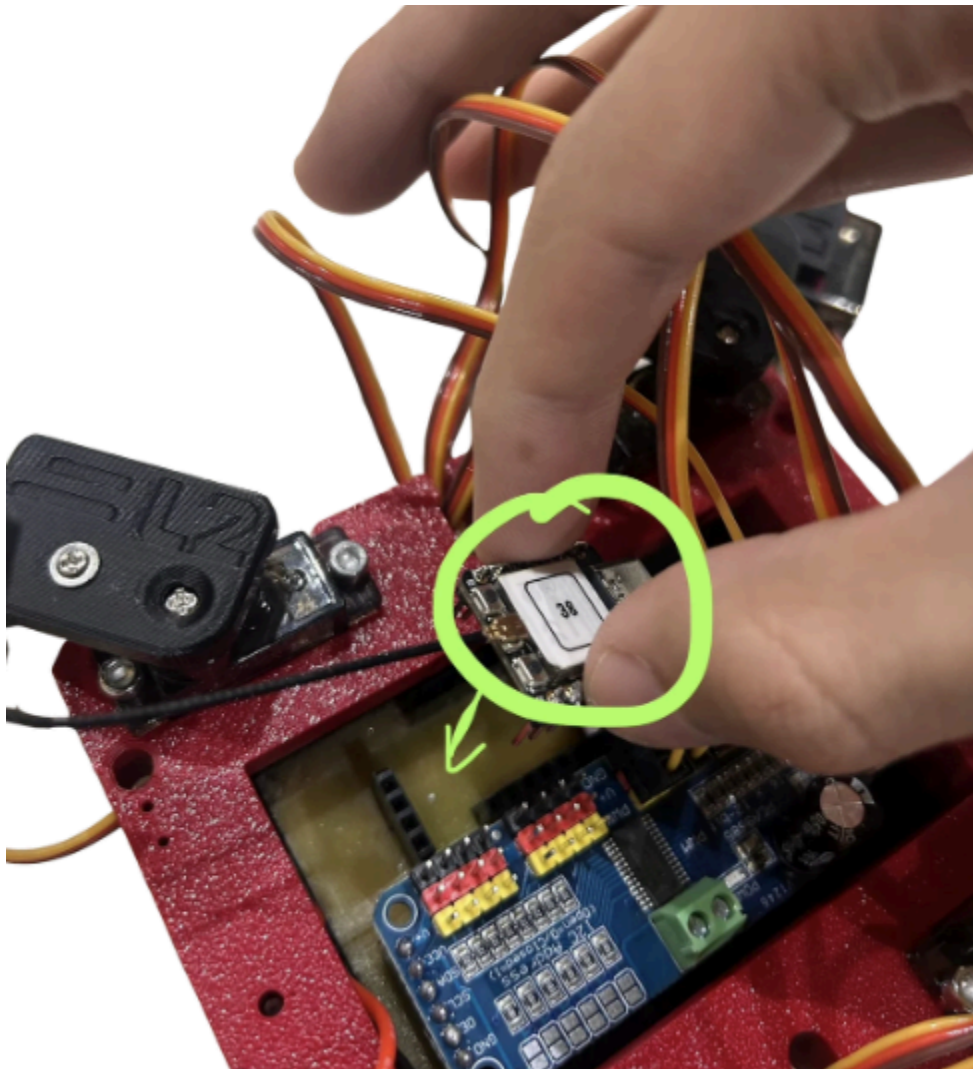
**Step 10:** After placing all the wires into the right pins, attach the Servo Driver to the PCB using the pins on the bottom of the Servo Driver, inserting it into the slots present in the PCB.





**Step 11:** Now, do the same for the ESP 32.

**Important:** Place the ESP in such a way that the USB C port faces the hole in the bottom plate.



Now your Ronnie should look like the one depicted below!

